

GLOBAL INDUSTRY CHALLENGE 2026

TEAM COVER PAGE

Conditional Generative Quantum Eigensolver for Efficient EUV Materials Simulation 19/04/2025				
Team Name: Ryoushi (Quantum Buddies)				
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Team Qualifications

Gyanateet Dutta (Ryukijano) – MSc Computer Science & AI, University of Leeds. Research Scientist Intern with expertise in foundational self supervised learning AI-For-Sciences, hybrid quantum-classical workflows, VQE/ADAPT-VQE, and foundation model integration for quantum chemistry. Winner, Yale Quantum Hackathon 2025 (1st place); Grand Prix, Bradford Quantum Hackathon 2025 (quantum genomics QNLP). Dat Chi (Ryan) Le (ryancdle) – MSc Theoretical Physics, University of Sheffield | Sheffield Quantum Centre. Specializes in photonic quantum machine learning, generative modelling, and quantum feature maps. Quantum Computing Intern, UCL. Co-winner, Bradford Quantum Hackathon 2025 (Grand Prix, Healthcare). Sid Iliyasu (SuperPenguin) – Imperial College London & City of London. Expertise in quantum algorithms, hybrid quantum-classical systems, and optimization. Co-winner, Bradford Quantum Hackathon 2025 (Grand Prix, Healthcare track).

Solution Steps

- Curate small EUV-relevant Hamiltonians (H_2 , LiH, halogenated aromatics); compute DFT / CCSD(T) / FCI baselines.
- Train encoder-decoder transformer on (H, U, E) pairs from exact diagonalization: learn $p(U|H)$ — no RL, no per-instance optimization.
- Evaluate generalization: held-out molecules and geometries; benchmark energy error vs FCI/CCSD(T), circuit depth, gate count vs VQE.
- Position c-GQE as a high-fidelity oracle in a tiered ML workflow (generative ML $\rightarrow$ DFT $\rightarrow$ c-GQE)

Technical Approach & Quantum Advantage

Core Innovation: We treat quantum simulation as a **learned compiler from Hamiltonians to circuits**, $H(x) \rightarrow UH(x) \rightarrow U$, enabling transferable priors across chemical families.

Unlike VQE (per-molecule gradient optimization, barren plateaus), c-GQE generates circuits as discrete token sequences from a pre-trained transformer, eliminating quantum gradients and enabling **amortized** inference and zero-shot transfer.

Why Quantum? EUV photoresists (halogenated/tin-based systems, 13.5 nm) require accurate excited-state spectra (~ 92 eV) with strong correlation and multi-reference character where DFT fails and CCSD(T) breaks down. Quantum eigensolvers natively represent multi-determinant wavefunctions.

Phase 1 Scope: 4–12 qubit Hamiltonians (classically simulatable), supervised pretraining only.

Evaluation: energy error vs FCI, circuit efficiency, and generalization to unseen systems.

Projected Industry Impact

Learning reusable circuit priors enables amortized quantum simulation across chemical design spaces, reducing quantum hardware cost. Integration into materials informatics pipelines (generative ML $\rightarrow$ classical filters $\rightarrow$ c-GQE) enables scalable screening of EUV photoresists, aligning with Mitsubishi Chemical & AIST quantum-AI initiatives.

References

- [1] Nakaji et al. (2024). Generative Quantum Eigensolver. arXiv:2401.09253.
- [2] Minami et al. (2025). Conditional GQE. Digital Discovery, RSC.
- [3] Kharazi et al. (2026). Quantum simulations for EUV photolithography. arXiv:2602.20234.

